Key Concepts: Hypothalamic and Pituitary Hormones

1. The adenohypophysis contains five cell types that produce six hormones: ACTH, TSH, FSH, LH, GH, and PRL.

	Primary hypothalamic regulation	Tropic hormone secreted	Target endocrine gland	Peripheral hormone involved in negative feedback
Corticotrope	Corticotropin-releasing hormone (CRH)- stimulatory	Adrenocorticotropic hormone (ACTH)- protein	Zona fasciculata and zona reticularis of adrenal cortex	Cortisol
Thyrotrope	Thyrotropin-releasing hormone (TRH)- stimulatory	Thyroid-stimulating hormone (TSH)- glycoprotein	Thyroid epithelium	Triiodothyronine (T3)
Gonadotrope	Gonadotropin-releasing hormone (GnRH)- stimulatory	Follicle-stimulating hormone (FSH) and luteinizing hormone (LH)- glycoproteins	Ovary (theca and granulosa) Testis (Leydig and Sertoli cells)	Estrogen[a], progesterone, testosterone, inhibin[b]
Somatotrope	Growth hormone-releasing hormone (GHRH)- stimulatory Somatostatin- inhibitory	Growth hormone (GH)- protein	Liver, and also direct actions especially metabolic effects	IGF-1, GH (short loop)
Lactotrope	Dopamine- inhibitory	Prolactin (PRL)- protein	No target endocrine gland	None

[a] estrogen can also have a positive feedback in females
[b] inhibin selectively inhibits release of FSH from the gonadotrope

2. In endocrine axis regulation, long loop negative feedback to anterior pituitary and hypothalamus is provided by levels of the free hormone that is secreted by the peripheral endocrine gland. Short loop negative feedback to the hypothalamus is provided by levels of the stimulating hormone.
3. The hypothalamus-pituitary-adrenal axis includes CRH → ACTH → Cortisol. This axis is stimulated by many forms of stress. It shows a pronouced diurnal pattern with a peak in early morning.
4. The hypothalamus-pituitary-thyroid axis includes TRH → TSH → Thyroid Hormone. This axis is downregulated by stress and negative feedback comes primarily from biologically active T3.
5. The hypothalamus-pituitary-gonad axis includes GnRH → FSH and LH → Testosterone, Estrogen, Progesterone, Inhibin. Notably, GnRH is released in a pulsatile pattern. In this axis, the sex steroids provide negative feedback.
6. The hypothalamus-pituitary-liver axis includes GHRH → GH → IGF-1. Both GH and IGF-1 have systemic physiological effects. Somatostatin is a peptide that is released from the hypothalamus and functions as an inhibitory signal, including inhibition of somatotrophs, which prevents GH synthesis and release.
7. The hypothalamus-pituitary-mammary gland is not considered an axis because prolactin (PRL) acts directly on nonendocrine cells. PRL release from the anterior pituitary is normally inhibited by dopamine from the hypothalamus. PRL promotes release of dopamine.
8. Endocrine diseases are classified as 1°, 2°, or 3° based on the level of lesion. Lesion level can be inferred by examining relative levels of releasing, tropic, and peripheral hormones involved in the axis.
9. The neurohypophysis (posterior pituitary) produces two hormones: oxytocin and ADH.
10. Oxytocin is regulated by myometrial contractions (positive feedback loop) and suckling during breastfeeding. Myoepithelial contractions are stimulated by GPCR signaling through IP_3.
11. ADH is regulated by changes in plasma osmolality and hemodynamics. It enhances water absorption from tubular fluid in distal tubule and collecting duct. ADH dysregulation is implicated in multiple diseases, including central diabetes insipidus, syndrome of inappropriate ADH secretion, and nephrogenic diabetes insipidus.

Hypothalamic and Pituitary Hormones- Handout
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Objectives

1. Describe the anterior pituitary lobe with respect to histology, cell types, vascular supply, development, and anatomical function relative to the hypothalamus.
2. Discriminate between releasing and inhibitory factors from hypothalamic neurons and trophic hormones produced by cells of the adenohypophysis, and describe their route of transport from the hypothalamus to the anterior pituitary.
3. Describe the three major families of the anterior pituitary hormones and their effects on target organs.
4. Describe negative feedback control of anterior pituitary hormone secretion at the hypothalamic and pituitary levels.
5. Describe the posterior pituitary with respect to histology, cell types, vascular supply, development, and anatomical functions relative to the hypothalamus.
6. List the target organs, stimuli and functional effects of oxytocin.
7. Describe the stimuli and mechanisms that control vasopressin secretion.
8. List the target cells for vasopressin and explain why vasopressin is also known as antidiuretic hormone.
9. Identify primary, secondary and tertiary endocrine disease based on plasma levels of hypothalamic-releasing hormones, stimulating hormones, and target hormones.

Negative feedback decreases the deviation from an ideal normal value and is vital in maintaining homeostasis. Most endocrine glands are under the control of negative feedback. In this mechanism, a hormone acts on a target organ to induce a change (increase or decrease) in the circulating levels of a controlled parameter (i.e., glucose levels). Changes in this parameter inhibit the secretion of the hormone. Negative feedback mechanisms act like the thermostat that controls the temperature in the home.

Many of these mechanisms depend on factors that influence the metabolism and excretion of hormones. Thus, higher hormone concentration alone cannot trigger the negative feedback mechanism. Negative feedback must be triggered by overproduction of an "effect" of the hormone.

Positive feedback mechanisms control self-perpetuating events that can be out of control and do not require continuous adjustment. In positive feedback mechanisms, the original stimulus is promoted rather than inhibited, creating instability. Unlike negative feedback that maintains hormone levels within narrow ranges; positive feedback is rarely used to maintain homeostatic functions. Positive feedback increases the deviation from an ideal average value until an event occurs “something has to give.”

Childbirth is an example of positive feedback. During labor, the stretching of the cervix stimulates oxytocin release. Oxytocin stimulates and enhances labor contractions. As the baby moves toward the birth canal, pressure receptors within the cervix send messages to the brain to produce more oxytocin. Oxytocin travels to the uterus through the bloodstream, stimulating uterus contractions. The contractions intensify and increase until the baby is outside the birth canal. When the stimulus to the pressure receptors ends, oxytocin production stops and labor contractions cease.

Endocrine Axis Negative Feedback

Much of the endocrine system is organized into **endocrine axes**. Each axis consists of the hypothalamus, the pituitary, and a peripheral endocrine gland.

Hypothalamus - Pituitary

The hypothalamus is a region of the brain involved in endocrine, autonomic and behavioral functions; it also links the nervous system with the endocrine system. It contains several types of neurons responsible for secreting different hormones.

One of the most critical functions of the hypothalamus is to link the nervous system to the endocrine system via the pituitary gland (hypophysis) to maintain homeostasis. To achieve this task, the hypothalamus receives inputs about the state of the body and initiates compensatory changes if a determined homeostatic parameter drifts out of the normal range.

Afferent connections to the hypothalamus come from **Nucleus of solitary track, Reticular formation, and Retina.** The hypothalamus also has some intrinsic receptors including thermoreceptors and osmoreceptors.
Hypothalamus sends **neural signals to the autonomic system** and **endocrine signals** to/through the pituitary. Ultimately, the hypothalamus controls every endocrine gland in the body.
The pituitary is located below the hypothalamus. Communication between these two structures is crucial for the maintenance of the homeostasis and is essential for reproduction.

The hypothalamus contains several nuclei, each with a specific function.
If we make a transversal cut, we will see that the hypothalamus is split symmetrically into left and right halves by the 3rd ventricle (3V), which contains cerebrospinal fluid. The hypothalamus forms the floor and lateral walls of the 3rd ventricle. In these walls, there are clusters of neurons that have specific functions. Some of the more critical areas are Paraventricular (PVN), supraoptic (SO), ventromedial (VM) and arcuate nuclei.

Pituitary
Although the pituitary is connected to the hypothalamus, it is placed outside of the brain and sealed off by the "*diaphragma sellae*".
The pituitary is encased in the sphenoid bone forming a structure called the "*sella turcica*."
The hypophysis is composed of mainly two parts or pars:
1- The pars distalis or anterior pituitary or adenohypophysis.
2- The pars nervosa or posterior pituitary or neurohypophysis.
In the adenohypophysis, we can distinguish three parts:
- The distalis
- The pars tuberalis (a thin layer of cells), and
- The pars intermedia is located between the neuro- and the adenohypophysis and disappear in the adult.

The structure connecting the pituitary with the hypothalamus is called the pituitary stalk or infundibulum that connects the hypophysis to the overlying hypothalamus in the region of the median eminence.
Notice that the **adenohypophysis has epithelial components** and that the **neurohypophysis is a neural structure**.

Pituitary: Embryonic Origins
The pituitary has two embryologic origins:
1. Primitive brain
2. Epithelial cells of the oral ectoderm from the developing oral cavity

During development the primitive forebrain grows toward the oral cavity, this neural growth is called infundibulum. The infundibulum stimulates the growth of the epithelial cells of the roof of the oral cavity to grow. These cells grow to form what it is called the Rathke's pouch. This pouch of epithelial cells keeps growing until it reaches the infundibulum and at the same time, they lose their connection with the oral cavity.
The cells that are in contact with the infundibulum contribute to form the pars intermedia, and the rest contribute to the pars distalis.
At the end of the developmental period, the pituitary is formed by pars tuberalis, intermedia, and distalis forming the adenohypophysis, and the pars nervosa, which is an **expansion** of the infundibulum.
The infundibulum remains connected to the hypothalamus at the median eminence.

Hypothalamus-Pituitary Connections
There are two types of connections: **neural and vascular**.
Vascular Connection: the adenohypophysis communicates with the hypothalamus via a **vascular connection**, more specifically through a portal system of vessels. A portal system begins with capillaries and end in capillaries. In this case, the superior hypophyseal artery forms the first capillary web at the median eminence. These capillaries form new vessels that are directed to the adenohypophysis where they form the second capillary plexus. These form the **hypothalamic-hypophysial portal system**.
The hypothalamic-hypophysial portal system transports hormones produced by hypothalamic neurons. These hormones are secreted into the first portal plexus helped by open spaces in the capillaries wall that facilitates the transfer of hormone from the extracellular space to the capillary. These are called **fenestrated**

capillaries. Hormones secreted into these capillaries are transported to the adenohypophysis where they will diffuse out and stimulate specific endocrine cells located in the **adenohypophysis**. These cells, in turn, will secrete a second hormone directly into the general circulation.

The **neural connection** is established between the hypothalamus and the neurohypophysis. Notably, with two nuclei: the PVN and the SON. Neurons located in these nuclei project their axons all the way to the pituitary passing through the infundibulum. At the pars nervosa, the axons are in close contact with specialized cells called pituicytes, glial-like cells that provide support for the axon. This axon contains hormones that are produced by the neuron soma and transported to the terminal where they are stored waiting for the right stimulus to be secreted. These axons are also in close contact with capillaries coming from the inferior hypophyseal artery. The axons unload their hormones in these capillaries.

Pituitary Cell Types

The adenohypophysis is composed of five endocrine cell types supported in acini by folliculostellate cells.

- Somatotrophs synthesize growth hormone (GH), which regulates bone, muscle growth, and maintains lean body mass in adults.
- Lactotrophs make prolactin (PRL), which inhibits gonadal function and stimulates breast milk production during and after pregnancy.
- Corticotrophs make adrenocorticotropin (also known as adrenal corticotropic hormone or ACTH), which stimulates glucocorticoid production by the adrenal cortex.
- Thyrotrophs synthesize thyrotropin (also known as thyroid-stimulating hormone or TSH), which stimulates the thyroid gland to produce thyroid hormones.
- Gonadotrophs produce follicle-stimulating hormone (FSH) and luteinizing hormone (LH), which regulate germ-cell development and sex steroid hormone production in the gonads.

Folliculostellate cells form a three-dimensional network, supporting and modulating endocrine cells by intercellular communication in a paracrine manner. Folliculostellate cells also produce bioactive peptides including basic fibroblast growth factor, vascular epithelial growth factor, IL-6, and neuronal nitric oxide synthesis. Pituitary tumors consisting only of folliculostellate cells have been described.

Hypothalamus – Pituitary - Peripheral Glands AXIS

Hormones produced by the posterior pituitary or neurohypophysis.

Two hormones are produced by neurons found in the PVN and SON nucleus. These are **Antidiuretic Hormone (ADH)** (also known as Vasopressin) and **Oxytocin**. Both hormones are nine amino acid peptides and have very similar structure differing only by two amino acids.

Vasopressin and Oxytocin are produced in **Magnocellular Cells of the SON and PVN**. These hormones are synthesized as pre-pro-hormone. The signal peptide is then cleaved in the endoplasmic reticulum and then packaged and transported to the **axon terminal found in the Pars Nervosa**. During transport, the prohormone is cleaved into ADH or Oxytocin and a neurophysin. Neurophysin is not active. The hormones are stored at the axon terminal in secretory granules also known as **Herring Bodies**.

When the Magnocellular cells receive the appropriate stimulus, they will send an action potential down the axon to the terminal leading to the fusion of the vesicle with the plasma membrane and the consequent secretion of the hormone into the general circulation. This process is facilitated by the presence of an extensive web of fenestrated capillaries in the posterior pituitary.

Oxytocin has two primary functions, both related to reproduction. This hormone induces the contraction of: **a)** the uterine myometrium during labor and **b)** of the myoepithelial cells surrounding the mammary gland alveoli during lactation.

In the case of labor, myometrial contractions stimulate the cervix. This signal is carried to the hypothalamus via the brainstem. This causes the secretion of oxytocin that in turn will increase contractions leading to parturition.

In the case of the mammary gland, the release of oxytocin is also mediated by a neuroendocrine reflex. In this case, the reflex is initiated by suckling, which stimulates sensory receptors in the nipple and areola of the breast. Consequently, there is activation of the magnocellular neurons and secretion of oxytocin.

Suckling also stimulates **parvocellular neurons** in the hypothalamus. Stimulation of these neurons leads to a decrease of dopamine secretion. Because dopamine inhibits prolactin secretion, the outcome is the stimulation of prolactin, which in turn stimulates milk production.

Oxytocin binds to its receptor on myoepithelial and myometrial cells and causes contraction through a Gαq-phospholipase C signaling pathway that ultimately increases intracellular calcium.